ScanIQ Livestock Embryo Intelligence: Commercialization Pitch Deck and Montana Ecosystem Partnership Strategy

Metrological Architecture and Technology Transition

Assisted Reproductive Technology (ART) within the commercial livestock sector operates under strict economic and biological realities that diverge sharply from human clinical in vitro fertilization (IVF). While human fertility clinics prioritize extensive genetic biopsies and high-cost manual interventions, commercial beef cattle and performance equine production operate on volume, speed, non-destructive workflows, and recipient herd efficiency. The conventional evaluation standard established by the International Embryo Transfer Society (IETS) relies on subjective, single-timepoint morphological grading under stereomicroscopy. This visual paradigm identifies cellular failure only after irreversible physical arrest or fragmentation has occurred. Every undetected non-viable embryo transferred into a surrogate cow or mare wastes an entire reproductive cycle, expensive synchronization pharmaceuticals, and critical production time.

The technical foundation for ScanIQ Montana LLC transitions the sensing architecture away from unvalidated human clinical claims to focus on an empirical biophysical metric: mammalian embryos generate a measurable electrical impedance shift near 1.49\text{ MHz} that directly correlates with expanding cellular mass and membrane surface area.

Prior iterations of this platform contained academic anchors that fail under commercial scrutiny:

- The projected 0.92 Area Under the Receiver Operating Characteristic (AUC) was a speculative human clinical benchmark rather than an empirical livestock dataset.
- The 13-stage progression model originated in teleost fish (*Oryzias latipes* / Medaka) experiments and does not map to mammalian ungulate cleavage stages.
- The 510 MPa flexural strength benchmark described the mechanical toughness of thermomechanically consolidated silk fibroin composites, a metric irrelevant to microfluidic dielectric sensing.
- Quantum magnetometry through nitrogen-vacancy (NV) diamond lattices and Schumann resonance (7.83 Hz) common-mode filtering introduced unnecessary complexity. Montana agricultural producers, seedstock operators, and technology boards require robust, solid-state electrical metrology rather than laboratory quantum magnetometers.

ScanIQ builds an automated culture cassette that interrogates embryos passively and continuously across high-frequency dielectric regimes. By combining electrical impedance spectroscopy (EIS) centered around the 1.49\text{ MHz} relaxation marker with high-frequency resonant frequency shifts (\Delta f_r) and quality factor (Q) metrics, the cassette monitors developmental trajectories without physical handling, dye exposure, or destructive radiation.

The geometry of the sensor is designed, printed, and iteratively refined using a field-native geometry engine built upon Signed Distance Fields (SDF) and volumetric material tensors via the PicoGK compiler and Helix Neural Network. This closed loop allows the additive

manufacturing system to test whether complex non-orientable topological geometries, such as single-sided Möbius strip resonators that lift clockwise/counter-clockwise polarization degeneracy, provide superior signal-to-noise ratios compared to standard micro-impedance electrodes.

The technology transition is supported by a synchronized operational strategy in Montana. By locating research, manufacturing, and validation within the Livingston–Bozeman–Yellowstone agricultural corridor, ScanIQ connects academic reproductive physiology at Montana State University (MSU) with commercial seedstock operators and regional non-dilutive grant funds.

Pitch Deck Presentation: ScanIQ Livestock Embryo Intelligence

Slide 1: Cover

- **Headline**: ScanIQ Livestock Embryo Intelligence
- **Sub**: Non-invasive viability scoring for bovine and equine embryo transfer, built on manufactured resonators and temporal AI.
- **Footer**: Montana ScanIQ Biotech Park proposal, September 2026.
- **Visual Direction**: Dark obsidian technical substrate overlaid with a 3D CAD cutaway of the dual-sided microfluidic culture well. Translucent dielectric channels show high-frequency field distribution contours around a central mammalian embryo trap.
- **Visual Caption**: [Concept Render]

Slide 2: The Problem

- **Headline**: Embryo selection in cattle and horses is still done by eye.
- **Body**: Grading is visual, subjective, and happens once. Failure is detected only after arrest. Every non-viable transfer costs a recipient cycle, synchronization drugs, and 30 to 60 days of a producer's calendar—totaling $450 to $650 per failed bovine recipient and over $2,200 per open recipient mare.
- **Visual Direction**: Split graphic contrasting an ambiguous, single-plane stereomicroscopic image of an early blastocyst alongside a timeline illustrating 60 lost production days, synchronization drug costs, and an open recipient cow in a chute.
- **Visual Caption**: [Concept Render]

Slide 3: What Exists Today

- **Headline**: The physics is published; the instrument isn't.
- **Body**: Bovine embryos show a measurable impedance shift near 1.49 MHz that tracks cell mass, IEEE EMBC 2024. Time-lapse imaging gives independent stage labels. Silk-based resonators shift with dielectric load. Nobody has fused these into a continuous, hands-off sensor for a culture dish. That's the gap.
- **Visual Direction**: Frequency response graph highlighting the differential real impedance curve (\Delta\text{Re}(Z)) with an inflection marker at 1.49 MHz, flanked by optical time-lapse stage progression thumbnails and a microfluidic well schematic.
- **Visual Caption**: [Concept Render]

Slide 4: What ScanIQ Builds

- **Headline**: A culture cassette that watches the embryo electrically, continuously, without touching it.
- **Body**: Impedance, resonance shift and Q, and optical morphokinetics, synchronized with incubator telemetry, feeding a temporal model that scores trajectory deviation before morphology changes. Passive measurement only. No stimulation.
- **Visual Direction**: Cross-sectional architectural illustration of the multi-well culture cassette detailing the non-contact sensing boundaries, microfluidic perfusion channels, and the sensor bus carrying RF, optical, and environmental data.
- **Visual Caption**: [Concept Render]

Slide 5: Why the Resonator Matters

- **Headline**: Manufactured geometry becomes the sensor.
- **Body**: ScanIQ's field-native kernel designs the resonator, prints it, measures the as-built response, and redesigns. The Möbius topology is the first geometry we test, chosen because it lifts polarization degeneracy; it is a hypothesis, and the sprint tests it against plain impedance electrodes. Say that plainly—reviewers reward it.
- **Visual Direction**: Structural geometry rendering of an additive-manufactured Möbius strip resonator illustrating its non-orientable, single-sided manifold and the geometric Pancharatnam-Berry phase twist.
- **Visual Caption**: [Concept Render]

Slide 6: Why Bovine First, Equine Second

- **Headline**: Bovine validates; equine differentiates.
- **Body**: Bovine has published anchors, high cycle volume, and Montana's herd. Equine has no published dielectric atlas, larger embryos, and high value per transfer. Bovine is the proof; equine is the first-mover dataset nobody else holds.
- **Visual Direction**: Comparative two-panel matrix displaying high-throughput bovine micro-wells alongside a larger, specialized equine culture capsule, highlighting cycle volumes, market values, and development milestones.
- **Visual Caption**: [Concept Render]

Slide 7: Why Montana

- **Headline**: The herd, the lab, and the customer are in the same state.
- **Body**: Animal-use oversight rather than human reproductive regulation. MSU animal science and veterinary research base (Department of Animal and Range Sciences, Montana Agricultural Experiment Station). Working ET operations as first customers and data partners. Ag-tech commercialization funds without dilution (Montana Growth Through Agriculture, Montana Board of Research and Commercialization Technology).
- **Visual Direction**: Regional Montana map connecting the research core at Bozeman (MSU) with commercial seedstock cattle ranches and equine veterinary surgical hospitals across the Yellowstone corridor.
- **Visual Caption**: [Concept Render]

Slide 8: The 12-Week Sprint

- **Headline**: One question, twelve weeks: does a printed resonator add signal beyond electrodes?
- **Body**: Weeks 1–4 materials and coupon library. Weeks 5–8 cassette and impedance acquisition. Weeks 9–12 blinded bovine run, fifty embryos, stage prediction against time-lapse labels. Exit gate: replicate the 1.49 MHz curve and beat electrodes-alone on held-out embryos, or stop.
- **Visual Direction**: Operational Gantt timeline divided into three 4-week phases, culminating in a clear binary pass/fail decision milestone.
- **Visual Caption**: [Concept Render]

Slide 9: Validation Roadmap

- **Headline**: Structured metrological and biological de-risking.
- **Body**: A five-phase progression establishing electrical reproducibility, non-living phantom calibration, multi-species validation, and on-ranch deployment.

Phase	Milestone	Validation Criteria
Phase 01	Bench Phantom	Resonator resolves calibrated dielectric and geometric fluidic perturbations.
Phase 02	Non-Living Models	Repeatable spectral signatures across synthetic microspheres and organoids.
Phase 03	Bovine, Then Equine	Track stage transitions, cell mass, and arrest under blinded conditions.
Phase 04	Multimodal Model	Sensor fusion outperforms optical imaging alone on held-out embryo cohorts.
Phase 05	Field Trials with Montana ET Operators	In-field viability scoring and recipient pregnancy confirmation with commercial partners.

- **Visual Direction**: Five-step horizontal progression card highlighting experimental models, testing methods, and pass/fail exit gates.
- **Visual Caption**: [Concept Render]

Slide 10: Business Model

- **Headline**: Cassette per transfer, software per score.
- **Body**: Consumable cassette ($85 bovine / $275 equine), recurring per-embryo scoring ($35 bovine / $75 equine), hardware placed with ET labs. A single avoided bovine recipient failure saves an operator $450 to $650, returning a 4x immediate return on investment on metrology fees.
- **Visual Direction**: Waterfall financial diagram demonstrating unit margins across

single-use cassettes, cloud algorithmic scoring tiers, and net cost savings per confirmed pregnancy for the producer.
- **Visual Caption**: [Concept Render]

Slide 11: Intellectual Property

- **Headline**: Four filings, one kernel.
- **Body**: Field-native resonator co-design; passive resonator sensing fixture; multimodal developmental phenotyping; closed-loop manufacture-to-measurement. Provisional filed before this deck circulates.
- **Visual Direction**: Central schematic showing the core PicoGK/Helix geometric kernel feeding into four patent application boundaries covering hardware, fabrication toolpaths, and predictive neural network weights.
- **Visual Caption**: [Concept Render]

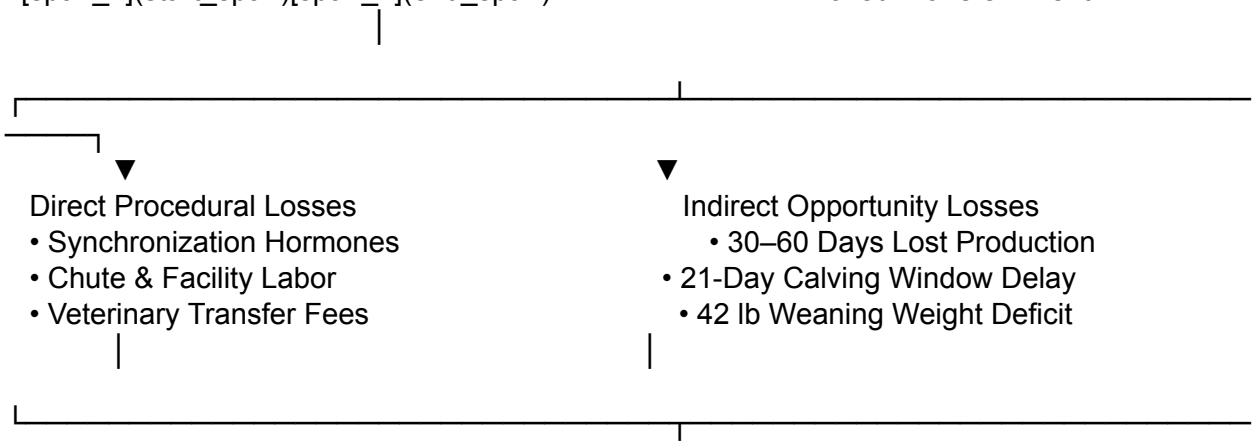
Slide 12: The Ask

- **Headline**: Lab space, a herd partner, and a sprint budget.
- **Body**: Park incubator space with incubator and impedance bench; MSU embryology collaborator; ET operator supplying embryos and outcomes; sprint funding of $165,000 (covering a $10,500 VNA setup, $25,000 benchtop incubator, $14,500 embryo run, and precision bench engineering).
- **Visual Direction**: Itemized capital budget allocation chart showing hardware procurement, facility wet-lab lease, biological supply, and systems engineering support.
- **Visual Caption**: [Concept Render]

Empirical Financial Analysis: Economics of Embryo Transfer Failure

Commercial livestock embryo transfer depends directly on recipient synchronization efficiency. In commercial artificial insemination (AI), an open cow costs an operator $46 to $84 in lost semen and handling. In contrast, an open recipient cow following embryo transfer loses the entire cost of donor superovulation, synchronization drugs, veterinary labor, and two cycles of production maintenance.

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Cumulative Financial Penalty
• Bovine Failure: $450 – $650
• Equine Failure: $1,500 – $3,500

Bovine Recipient Economics

In beef cattle seedstock and commercial recipient programs, preparing an open recipient female requires an intensive estrus synchronization protocol, such as a 7-day CO-Synch + CIDR program. Pharmaceutical inputs include prostaglandin (F_{2\alpha}), gonadotropin-releasing hormone (GnRH), and an intravaginal progesterone-releasing device, totaling $15 to $30 per head in baseline chemistry. Administering this protocol requires three separate passes through the working chute, generating $30 to $55 in handling labor, animal stress, and facility depreciation. Veterinary non-surgical transcervical embryo placement adds a professional service charge of $50 to $100 per transfer.

When an embryo arrests post-transfer due to undetected developmental defects, the recipient remains open. Pregnancy diagnosis via transrectal ultrasonography or blood pregnancy-specific protein B (PSPB) testing occurs between Day 28 and Day 35 post-transfer. Over this 30-day monitoring period, maintaining an open recipient cow on a Montana feed ration costs between $2.50 and $3.50 per day in feed, mineral supplementation, and yardage, generating $75 to $105 in direct holding expenses.

The indirect costs of an open cycle are often more severe. In beef production, a cow that fails to conceive during an initial transfer slips to a subsequent 21-day estrus cycle, delaying her calving window. A calf born 21 days late gains approximately 2.0 pounds per day of lactation; this delay equates to a 42-pound deficit in weaning weight in the fall. At average market prices of $3.00 per pound for feeder calves, this delay represents an additional $126 loss in calf value. In commercial recipient herds, such as those providing 60-day confirmed pregnant cows for $2,200 to $3,200 per head, a failed transfer wastes capital and delays delivery. Summing procedural fees, synchronization drugs, feed, and weaning deficits, **the total cost of a single failed bovine recipient transfer ranges from $450 to $650**.

Equine Recipient Economics

The financial consequences are higher in performance equine breeding. Unlike bovine recipients, mares cannot be synchronized in large groups using fixed-time hormonal protocols, requiring individual reproductive management. Because an equine embryo must be collected via uterine flush between Day 7 and Day 8 post-ovulation and transferred into a recipient mare that ovulated within +1 to –3 days relative to the donor, commercial equine ET centers maintain large, non-pregnant recipient herds.

Clients pay a non-refundable recipient mare synchronization reservation fee ranging from $600 to $1,500 per attempt. Veterinary uterine flushing, embryo identification, handling, and non-surgical transcervical transfer cost between $500 and $1,200 per cycle. Recipient mare boarding and daily monitoring range from $18 to $35 per day.

If an embryo fails to establish a pregnancy—confirmed via ultrasound at Day 14 to 16—the synchronization fee, professional labor, and two to three weeks of recipient board are

completely unrecoverable. This sets the baseline **direct financial loss for a single failed equine transfer attempt at $1,500 to $3,500**. If an operator leases a recipient mare through weaning, full lease contracts range from $3,000 to $7,000, payable upon confirmed 35- to 45-day heartbeat detection.

Cost Category	Bovine Recipient Loss (30-Day Failure Cycle)	Equine Recipient Loss (16-Day Failure Cycle)
Synchronization Pharmaceuticals (GnRH, PGF_{2\alpha}, CIDR / Deslorelin, Altrenogest)	$15 – $30	$150 – $350
Handling, Labor, and Chute Processing	$30 – $55	$200 – $400
Professional Veterinary Transfer Fee	$50 – $100	$350 – $650
Recipient Herd Reservation / Chute Entry Fee	N/A (Internal Herd)	$600 – $1,500
Open Recipient Maintenance and Daily Boarding	$75 – $105 (30 days @ $2.50–$3.50/day)	$288 – $560 (16 days @ $18–$35/day)
Ultrasound Pregnancy Confirmation Exam	$15 – $30	$75 – $150
Calving Window Delay / Weaning Opportunity Loss	$126 – $210 (21-day cycle slip)	Seasonal breeding slip (High value)
Total Net Economic Loss per Failed Transfer	**$450 – $650**	**$1,500 – $3,500**

Academic and Commercial Partnership Framework in Montana

Developing, calibrating, and commercializing this platform requires access to live biological specimens, validated stage labeling, and field placement. ScanIQ leverages Montana's agricultural infrastructure by integrating academic researchers with commercial livestock operations.

Montana State University (MSU)
• Dr. Sarah McCoski (Bovine IVP Biology)
• Dr. Amanda Bradbery (Equine Reproduction)
• Spectrum Laboratory (RF & Test Metrology)



ScanIQ Montana LLC
(Additive Manufacturing & Sensor Core)



Commercial ET Field Operators
• Frontier Embryo Center (Recipient Herds)
• Sitz Angus (Registered Beef Seedstock)
• ORIgen Embryo Technologies (IVF Lab)
• Montana Equine (Equine ET Recipient Herd)

Academic Partner: Montana State University (Bozeman)

Montana State University is the state's land-grant research institution, housing the College of Agriculture and the Montana Agricultural Experiment Station (MAES).
The MSU Department of Animal and Range Sciences, located within the Animal Bioscience Building, provides research infrastructure in ungulate reproductive physiology, developmental biology, and gamete cryobiology. ScanIQ targets two specific academic investigators:

- **Dr. Sarah McCoski (Assistant Professor of Reproductive Biology)**: Dr. McCoski's laboratory studies ruminant embryogenesis, conceptus elongation, placentation, and developmental programming. Her research evaluates how early in vitro culture environments alter blastocyst gene expression, metabolic competence, and pregnancy retention in beef cattle. Her group provides the embryological oversight required to handle live bovine oocytes and zygotes, supply standardized IVP media, and provide blinded, independent morphological staging against time-lapse imaging.
- **Dr. Amanda Bradbery (Associate Professor of Equine Science & Reproductive Physiology)**: Dr. Bradbery's research investigates nutritional modulation, gestational environments, and maternal-fetal interactions in horses. Her ongoing work, funded in part by the Bair Ranch Foundation, examines the uterine microbiome and physiological determinants of reproductive success in mares. Collaborating with Dr. Bradbery gives the platform access to equine reproductive cohorts and clinical expertise to establish the industry's first equine embryonic dielectric atlas.
- **MSU Spectrum Laboratory**: For high-frequency microwave and electromagnetic characterization, the Spectrum Lab offers local facilities to test custom transmission lines, microstrip coupling, and vector network analysis up to multi-gigahertz bands, assisting in coupon library verification.

Commercial Embryo Transfer Partners

Field validation requires working livestock operations that process hundreds of donor flushes and recipient transfers each breeding season.

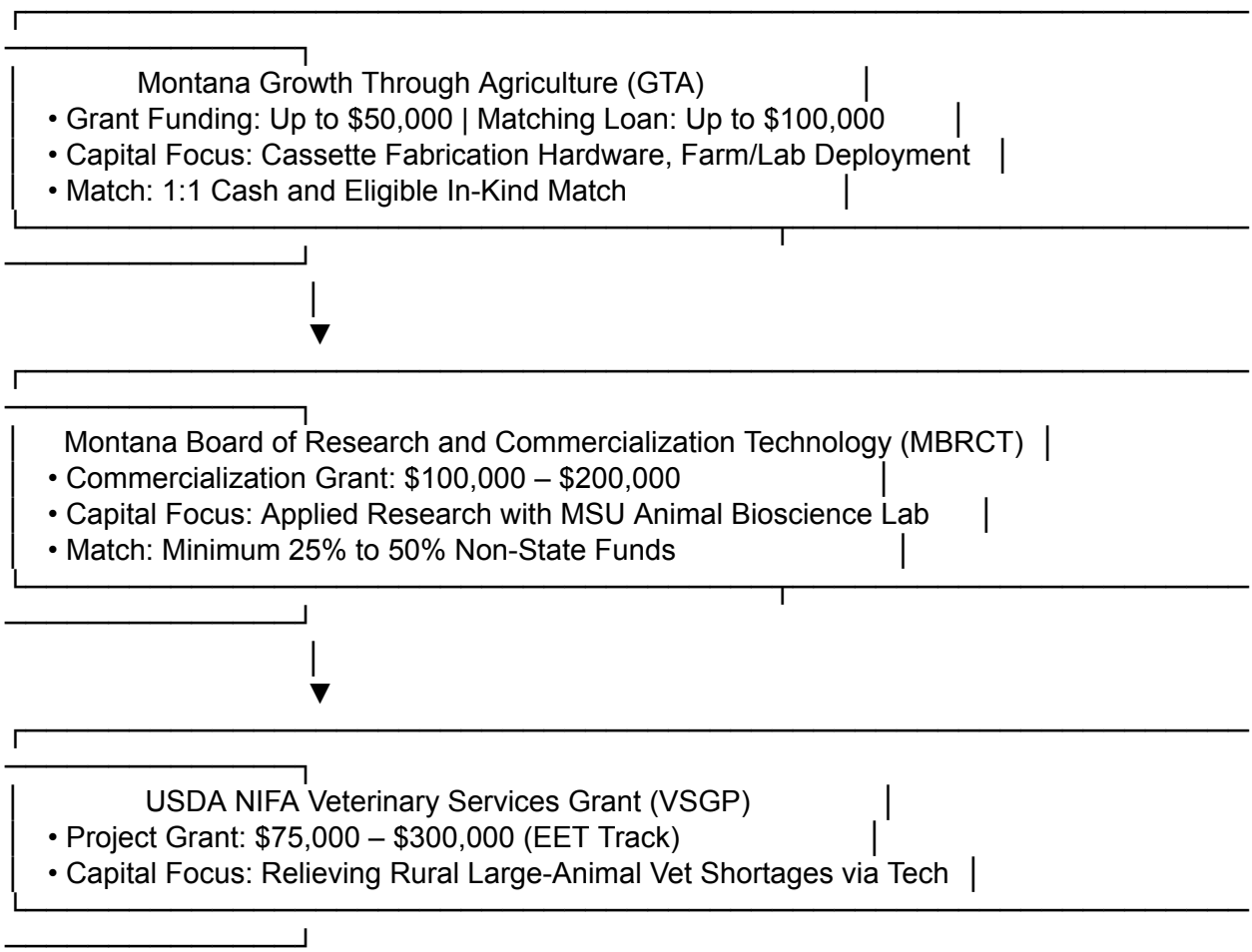
- **Frontier Embryo Center (Caprock Cattle Co., Lewistown, MT)**: Led by Jonathan Dietrich, Frontier Embryo Center manages large-scale commercial bovine recipient herds, specializing in 60-day confirmed pregnant recipient cows, conventional flush recovery, and in vitro fertilization (IVF) transfers. Their operation transfers hundreds of embryos annually into synchronized commercial recipient cows, providing an ideal operational environment for on-ranch field trials.
- **Sitz Angus (Harrison & Dillon, MT)**: Operated by Bob and Jim Sitz, Sitz Angus is a foundational American registered Angus seedstock enterprise. They run intensive maternal breeding programs using artificial insemination, genomics, and commercial embryo transfer to market hundreds of registered bulls and females annually. Sitz Angus serves as an enterprise reference customer, evaluating whether pre-transfer dielectric scoring improves first-service conception rates across commercial recipient herds.
- **ORIgen Embryo Technologies (Huntley / Billings, MT)**: ORIgen operates an advanced livestock reproductive facility offering transvaginal oocyte aspiration (OPU), conventional IVF, and proprietary serum-free embryo culture. Because ORIgen evaluates media chemistries and produces hundreds of direct-thaw embryos for cattle producers, they

provide the throughput needed to validate the microfluidic cassette against commercial standards.

- **Montana Equine Medical & Surgical Center (Three Forks / Gallatin Valley, MT)**: Montana Equine provides full-service equine reproductive care, including donor mare management, non-surgical uterine flushing, and a dedicated recipient mare herd. Partnering with Montana Equine allows ScanIQ to access clinical equine flushes, test Day-7 to Day-8 blastocysts within the culture cassette, and benchmark viability scores against ultrasound pregnancy confirmations.

Non-Dilutive Capitalization and State Grant Strategy

Montana maintains state programs designed to foster technology commercialization, strengthen agricultural value chains, and retain intellectual property within the state. These programs provide non-dilutive capital to fund early prototyping and validation sprints without prematurely diluting equity.



Montana Growth Through Agriculture (GTA) Program

The Montana Growth Through Agriculture program, administered by the Montana Department of

Agriculture and the Agriculture Development Council, offers grants up to $50,000 and low-interest matching loans up to $100,000. GTA funds target projects that add value to Montana agricultural commodities, introduce innovative technologies to rural operations, and establish local manufacturing.
ScanIQ fits GTA funding objectives:

- **Rural Economic Impact**: Establishing additive manufacturing and sensor assembly in the Livingston–Lewistown area creates skilled technical jobs and strengthens local supply chains.
- **Livestock Value Creation**: Improving embryo transfer success rates reduces waste for Montana seedstock producers, directly lowering breeding expenses.
- **Match Structuring**: GTA awards require a 1:1 capital match, which ScanIQ fulfills through founder capital, equipment contributions, and regional angel syndication.

Montana Board of Research and Commercialization Technology (MBRCT)

The Montana Board of Research and Commercialization Technology, managed under the Montana Department of Commerce, invests in applied technology projects with high commercialization potential. Historically providing awards between $100,000 and $200,000, MBRCT encourages joint ventures between private startup enterprises and university researchers across the Montana University System.
The program requires a minimum non-state matching fund contribution of 25% to 50% of the total project budget. A joint MBRCT proposal structured alongside the MSU Department of Animal and Range Sciences directly subsidizes laboratory testing, engineering bench fees, and animal care expenses for the multi-embryo validation run.

Complementary Federal Opportunities

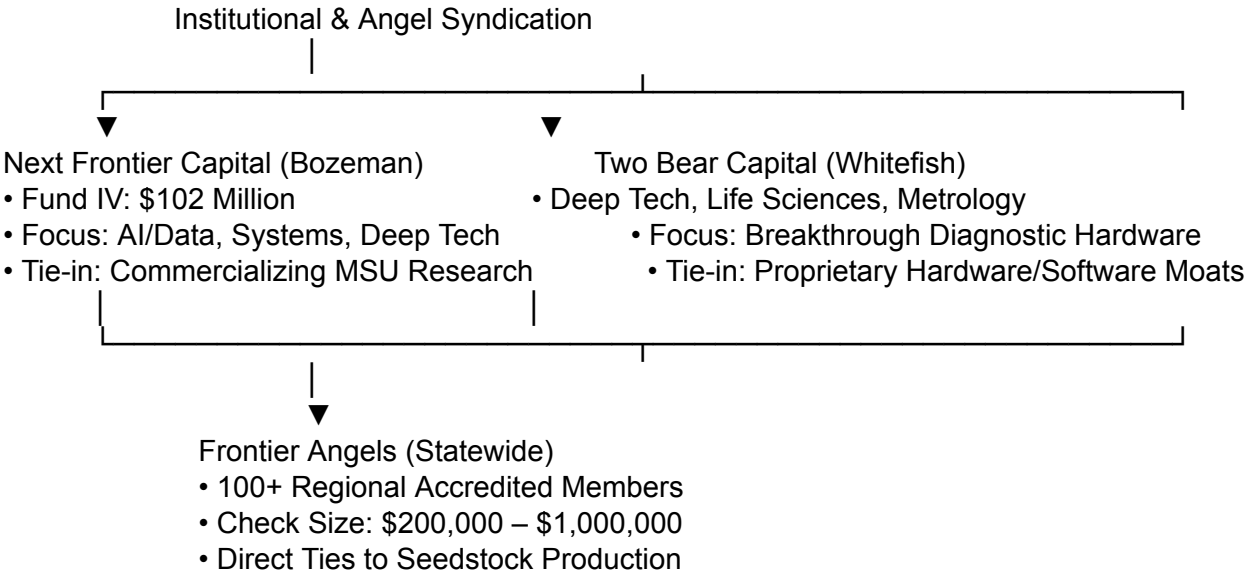
- **USDA NIFA Veterinary Services Grant Program (VSGP)**: Awards $75,000 to $300,000 under the Education, Extension, and Training (EET) track. This program addresses large-animal veterinary shortages across Montana by funding automated diagnostic systems that improve procedural efficiency in rural practices.
- **USDA Value-Added Producer Grant (VAPG)**: Provides $50,000 for planning or $250,000 for working capital (1:1 match) to assist agricultural producers and aggregators scaling technologies that enhance livestock value.

Funding Program	Administering Agency	Award Capital	Match Structure	Operational Allocation for ScanIQ
Growth Through Agriculture (GTA)	Montana Dept. of Agriculture	Up to $50k grant; up to $100k loan	1:1 cash/in-kind match	Cassette 3D printing equipment, resin synthesis, pilot testing.
Board of Research & Commercialization (MBRCT)	Montana Dept. of Commerce	$100,000 – $200,000	25% to 50% non-state match	Joint MSU wet-lab study, embryology staffing, and data acquisition.

Funding Program	Administering Agency	Award Capital	Match Structure	Operational Allocation for ScanIQ
USDA NIFA VSGP (EET Track)	USDA NIFA	$75,000 – $300,000	Program-specific / zero mandatory	Automated reproductive diagnostics for rural livestock veterinarians.
USDA Rural Development (VAPG)	USDA Rural Development	$50k planning; $250k capital	1:1 match (cash or eligible inventory)	Scaling commercial distribution across Montana seedstock operations.

Venture Capital and Regional Investor Syndicates

Montana's investment ecosystem features venture capital firms, angel networks, and family offices experienced in backing applied physical science, computational modeling, and agricultural ventures.



Next Frontier Capital (Bozeman, MT)

Next Frontier Capital (NFC) is the state's primary venture capital firm, managing $102 million in commitments through NFC Fund IV. NFC leads Seed and Series A investments ($2 million to $5 million check sizes) in deep-tech, artificial intelligence, and enterprise software startups located throughout the Rocky Mountain West.

- **Academic and Tech-Transfer Synergy**: General Partner Les Craig previously served as Executive Director of the Montana State University Innovation Campus, where he oversaw the creation of Bozeman's 42-acre Technology Park and launched the 406 Labs accelerator. NFC actively evaluates hardware-software platforms originating from regional

university research that establish defensible intellectual property moats.

Two Bear Capital (Whitefish, MT)

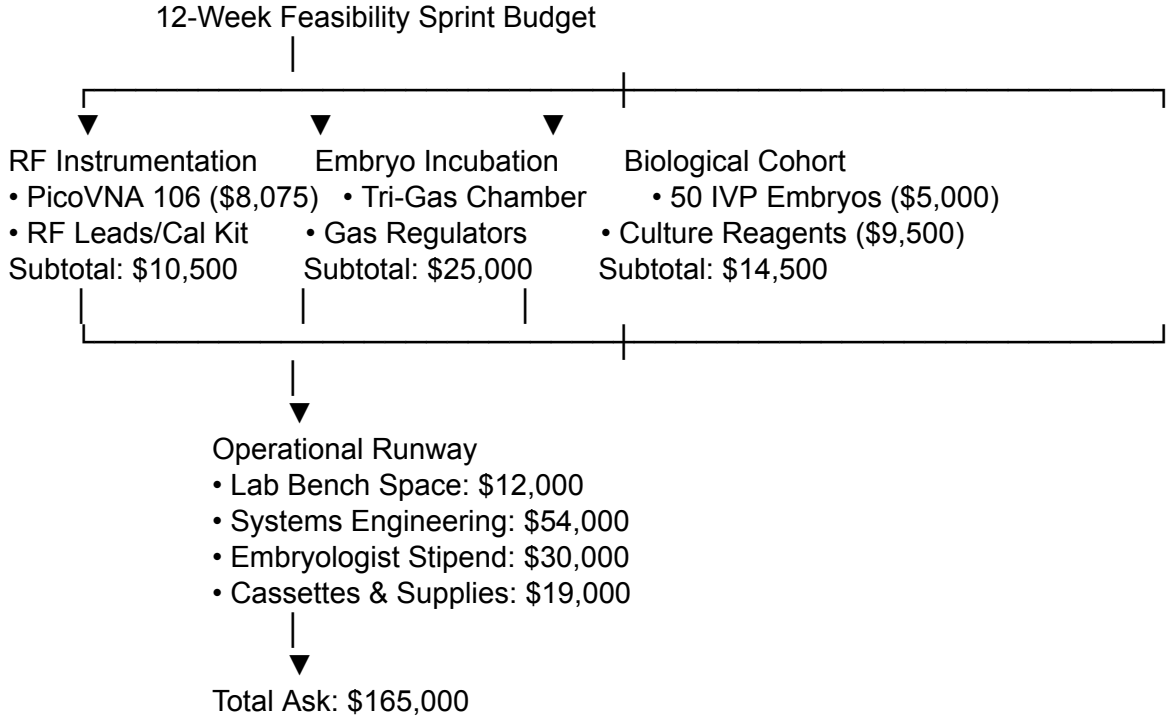
Two Bear Capital, founded by Michael Goguen, invests across biotechnology, bioinformatics, life sciences, and machine learning architectures. Liz Marchi, Head of Community Engagement at Two Bear Capital, was the founder of the Frontier Angel Fund network and has built regional early-stage bioscience funding networks. Two Bear Capital focuses on platforms that bridge physical metrology with machine learning to automate complex diagnostic tasks.

Frontier Angels (Statewide Montana)

Frontier Angels is an angel syndicate of more than 100 accredited investors across chapters in Bozeman, Billings, Missoula, and Kalispell. Frontier Angels participates in early-stage rounds between $200,000 and $1,000,000. Its membership includes ranch owners, large animal veterinarians, and commercial feedlot operators who understand the financial impact of embryo transfer failure, providing strategic domain expertise alongside early capital.

Sprint Sourcing, Instrumentation Budget, and Technical Exit Gates

The 12-week feasibility sprint addresses a single technical hypothesis: *Does a 3D-printed resonator provide measurable developmental signals beyond plain impedance electrodes, while replicating the published 1.49 MHz curve under blinded conditions?*
To run a capital-efficient program, the sprint deploys standard, commercially available RF and incubation hardware to establish an automated benchtop metrology workstation.



Vector Network Analyzer (VNA) Architecture

Measuring complex impedance (Z(f)), resonance shifts (\Delta f_r), phase, and loaded quality factors (Q) requires continuous, multi-frequency sweeps across the 100 kHz to 50 MHz band.

- **Base Instrument**: Pico Technology PicoVNA 106 (dual-port, 300 kHz to 6 GHz operation, 118 dB dynamic range, USB interface).
- **Base Price**: $8,075.
- **Calibration & Cabling**: Phase-stable 8.5 GHz flex-form SMA test cables (TA340/TA341), female SMA adaptors, and an 8.5 GHz SOLT (Short-Open-Load-Through) precision calibration kit (TA343) add $2,425.
- **Total RF Bench Cost**: **$10,500**.

Tri-Gas Embryo Incubation Chamber

Culturing mammalian ungulate embryos ex vivo requires tight physiological regulation: 38.5°C temperature, high relative humidity, and tri-gas balance (5\%\,\text{O}_2, 5\%\,\text{CO}_2, 90\%\,\text{N}_2) to prevent cellular arrest.

- **System**: Benchtop multi-room incubator (such as the Esco MIRI Multi-Room or Cook Medical MINC chamber).
- **Specifications**: Individual heated wells with fast parameter recovery (<1 minute temperature, <3 minutes gas) and on-board data logging to synchronize environmental telemetry with impedance streams.
- **Total Incubation Cost**: **$25,000**.

Biological Cohort (50-Embryo Blinded Bovine Run)

- **Embryo Supply**: 50 research-grade in vitro produced (IVP) bovine embryos sourced via abattoir-derived oocytes, fertilized with sex-sorted semen, and cultured in synthetic oviduct fluid.
- **Unit Cost**: Approximately $100 per viable blastocyst/transferable unit ($5,000 total).
- **Culture Consumables**: Culture media, oil overlays, microfluidic polymer resins, and fluidic tubing add $9,500.
- **Total Biological Supply Cost**: **$14,500**.

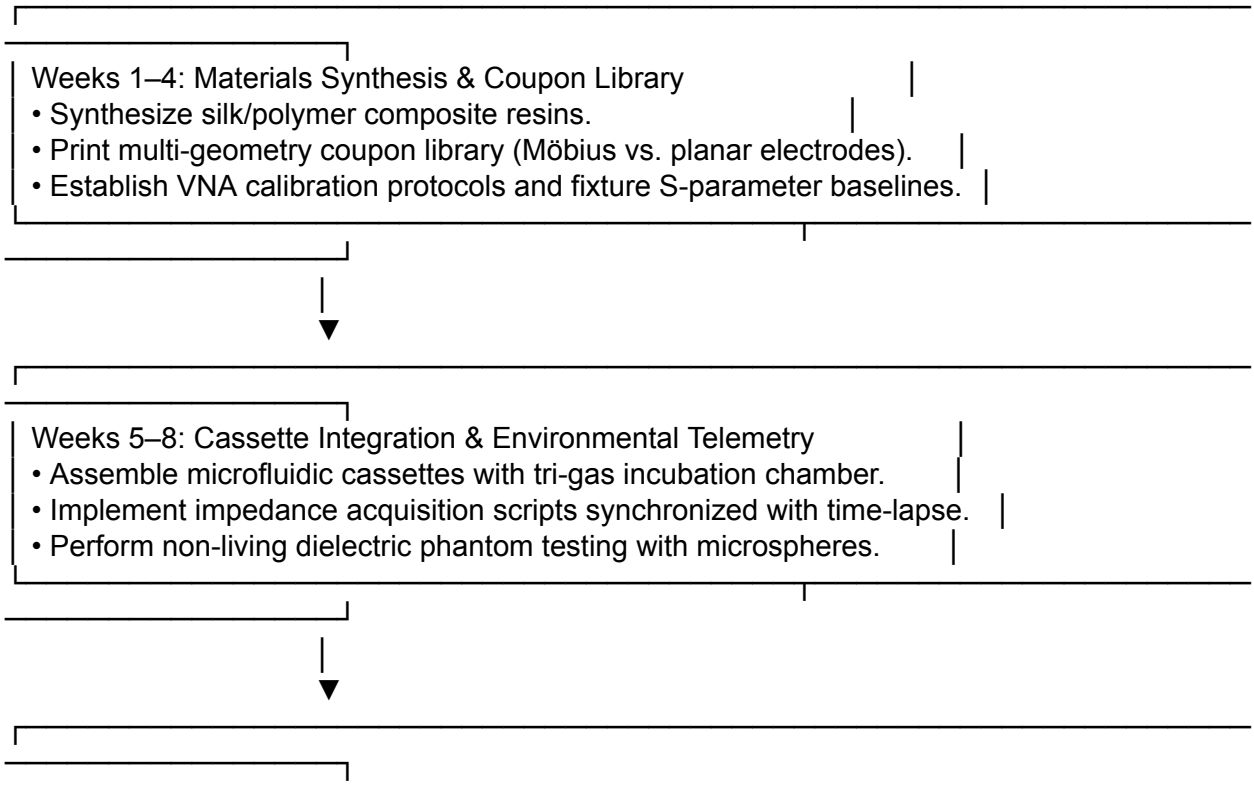
Sprint Budget Breakdown

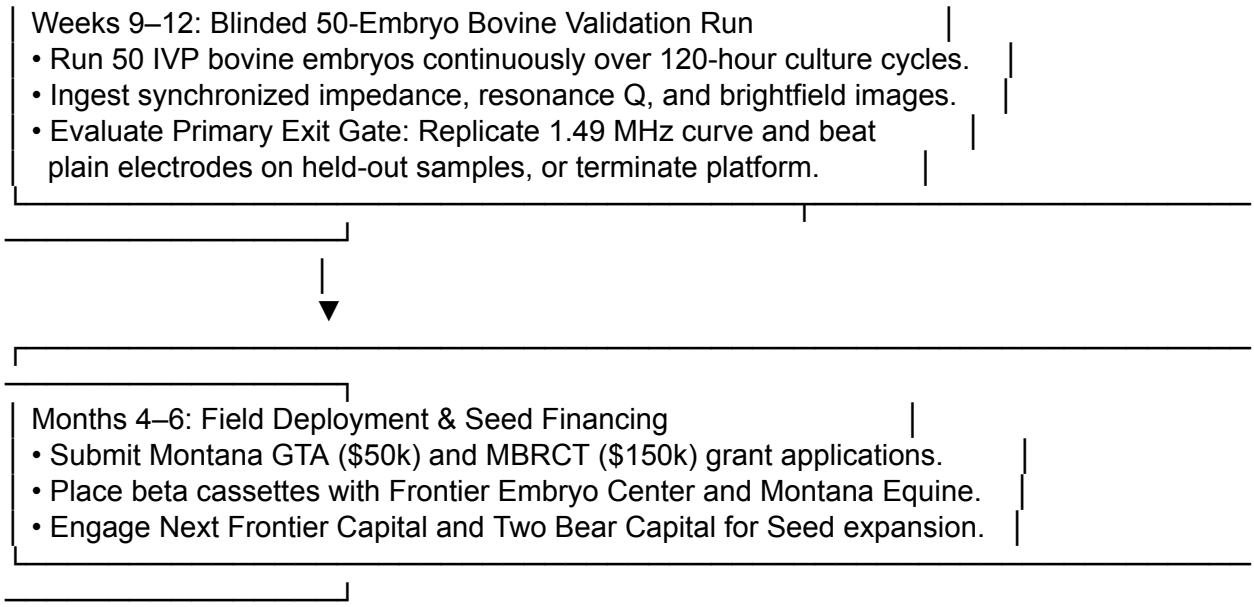
Expense Category	Itemized Description & Sourcing	Projected Cost (USD)
RF Metrology Bench	PicoVNA 106 (300 kHz – 6 GHz), phase-stable leads, SOLT cal kit.	$10,500
Tri-Gas Incubation	Esco MIRI / MINC benchtop incubator with gas mixing and data telemetry.	$25,000
Biological Supply	50 IVP bovine embryos,	$14,500

Expense Category	Itemized Description & Sourcing	Projected Cost (USD)
	synthetic oviduct media, and chemical reagents.	
Tooling & Prototyping	Photopolymer resins, cyclo-olefin substrates, and cassette fluidic fittings.	$9,000
Wet-Lab Lease	3 months shared wet-lab bench lease within regional biotech corridor.	$12,000
Embryology Specialist	Contract embryologist for specimen handling and independent labeling.	$30,000
Systems & RF Engineering	Antaramian / Systems engineering, PicoGK slicing, and VNA integration.	$54,000
Contingency / Calibration	Consumable replacements, gas refills, calibration checks, and travel.	$10,000
Total Sprint Budget	**Capital ask for 12-week feasibility demonstration**	**$165,000**

Strategic Commercialization Timeline

The ScanIQ roadmap progresses through discrete technical and operational gates, ensuring validation before capital expenditure increases.





The 12-week sprint focuses on testing the primary physical hypothesis. During Weeks 1 through 4, the engineering team prints silk-polymer composite test coupons using PicoGK implicit modeling, establishing clean manufacturing parameters and characterizing high-frequency dielectric losses. Weeks 5 through 8 integrate these coupons into fluidic cassettes mounted within the benchtop tri-gas incubator, verifying that temperature and gas stability match standard culture conditions while ensuring RF transmission lines remain calibrated.

Weeks 9 through 12 execute the blinded biological validation run. Fifty bovine IVP embryos are cultured individually within the cassettes across a standard 120-hour developmental window. The system continuously captures complex impedance spectra Z(f), loaded quality factors Q, and brightfield optical frames, streaming data into the temporal neural network.

The technical pass/fail criteria are strictly defined:

1. Replicate the published 1.49\text{ MHz} differential real impedance inflection tracking cell mass proliferation.
2. Demonstrate that the 3D-printed Möbius resonator extracts developmental trajectory shifts earlier than planar micro-impedance electrodes alone.
3. If the resonator fails to provide superior predictive signal on held-out samples, the sprint terminates.

Upon clearing this technical exit gate, Months 4 through 6 expand regional operations. ScanIQ leverages empirical sprint data to finalize a $50,000 Growth Through Agriculture grant and submit a joint $150,000 commercialization application with MSU to the Montana Board of Research and Commercialization Technology.

Simultaneously, beta sensing cassettes will deploy to commercial ET facilities—including Frontier Embryo Center and Montana Equine—to score embryos intended for transfer into recipient females. Correlating pre-transfer viability scores with 30-day ultrasound pregnancy confirmations provides the real-world validation needed to engage Next Frontier Capital, Two Bear Capital, and Frontier Angels for Seed funding, establishing Montana as a hub for livestock reproductive intelligence.

Works cited

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